Andrew Zheng

andrew.zheng1@uwaterloo.ca | in/andrewzheng2007 | github.com/awzheng | awzheng.me | Portfolio

Education

University of Waterloo | *BASc, Computer Engineering* | GPA: 3.9 — Sep 2025 – Apr 2030

Skills

Languages: C, C++, Python, Golang, Verilog, Java, SQL, HTML, CSS, JavaScript, TypeScript, Bash
DevOps/Tools: AWS (Lambda, S3, IAM), Docker, Git, Jupyter, Virtualization, VMWare Fusion, Linux
Libraries/Frameworks: FastAPI, React, MongoDB, Streamlit, NoSQL, REST API, MLOps, XGBoost, Tailwind CSS, Next.js
Embedded/Hardware: Altium Designer, KiCad, FPGA, STM32, ST-LINK, CAN, UART, I2C, SPI, Multimeter, Oscilloscope, Logic Analyzer

Experience

Software Engineering Intern, Device Management | Miovision | Waterloo, ON — May 2026 – Aug 2026

- Selected to engineer IoT infrastructure and distributed microservices for global traffic systems and 3 billion pedestrians

Embedded Systems Developer | Waterloo Rocketry | Waterloo, ON — Sep 2025 – Present

— **Remote Arming Board** | *Safety-critical Avionics Recovery PCB* — GitHub

- Engineered 4-layer KiCad PCB with PMOS high-side switching to arm remote recovery parachute via RocketCAN bus signal
- Architected hardware fail-safe via passive gate biasing to ensure dual-altimeter redundancy during MCU Hi-Z faults
- Implemented low-level **C** firmware for **PIC18** to bridge UART altimeter telemetry onto the RocketCAN bus via MCP2562

— **Fuel Injector Sensor Hub** | *Real-Time Propulsion Telemetry Firmware* — GitHub

- Developed **STM32** drivers in **C** to acquire pressure telemetry via ADC and log to RocketCAN and SD Card mid-flight
- Validated signal integrity and CAN/I2C communication protocol effectiveness using oscilloscopes and logic analyzers
- Validated and debugged hardware using NI automated test equipment to integrate custom EGSE for 2026 launch

Hardware Designer | UW Biomechatronics | Waterloo, ON — Sep 2025 – Jan 2026

— **EMG Bionic Arm** | *Bionic Arm PCB in KiCad and Altium Designer* — GitHub

- Spearheaded Altium migration and Git repository initialization to engineer **ESP32** control system with USB-C connectivity
- Verified PCB design constraints in Altium and SolidWorks to validate electromechanical integration for production

Case Competition Coach | Self-employed — Jun 2024 – Feb 2026

- Scaled self-founded coaching business from contract work to fully independent to achieve **5-figure revenue**
- Coached **150+** students from **10+** high schools to produce **50+** international qualifiers and **30+** international finalists

Projects

Allocate | Proactive macOS CPU Governor | *XNU Kernel, Mach IPC, Rust, Swift, POSIX* — GitHub

- Engineered native macOS QoS governor app to dynamically throttle background loads and sustain **100% P-core** availability
- Developed event-driven **Rust** daemon with thread-safe concurrency to process CPU/RAM telemetry with <**1ms** latency
- Architected SIP-compliant async subprocesses to force E-core migration and reduce default background power draw by **91%**
- Deployed sandboxed **SwiftUI** frontend via bidirectional **Mach XPC** to grant users real-time control of root CPU thresholds

SageWall | MLOps Cloud Security System | *AWS, Python, Streamlit, Machine Learning* — GitHub

- Architected IDS on **AWS SageMaker** (**XGBoost**) to analyze network traffic with **99.9%** accuracy and <**100ms** latency
- Automated Lambda ETL pipeline to transform **125,000+** NSL-KDD records in S3 and designed infrastructure using Boto3
- Deployed Streamlit web dashboard secured with AWS IAM, CloudWatch, and SNS for real-time system alerts

CrawlStars | Concurrent Search Engine | *Golang, JavaScript, MongoDB, REST API, Concurrency* — GitHub

- Engineered concurrent crawlers using **Golang** producer-consumer architecture to process **2000+ pages**/min in <**15MB**
- Optimized deduplication via sync.Map to enable O(1) lookup for **50,000+** URLs and to filter **70%** of redundant crawls
- Developed relevance algorithm using **MongoDB Atlas** aggregation pipelines to map SEO metrics to 5-star rating system

Mangaroo | PDF-to-Manga AI Illustrator | *FastAPI, Python, JavaScript, REST API, Gemini/Imagen API* — GitHub

- Architected async FastAPI for **10x** concurrent session handling to process **50MB** PDFs (**1000+** pages) in <**50ms**/page
- Engineered Story Bible state manager to reduce token usage by **97%** and cost from $24 to **$0.70** per 10K panels
- Deployed production-grade **3-layer** error architecture with singleton pattern and Vertex AI account authentication

Awards

1st Place World Champion (Marketing – Product Management) | *DECA ICDC 2024 in Los Angeles, California* — Apr 2024
7x National Honour Roll | *University of Waterloo CEMC Mathematics Contests* — 2020 – 2024